# Ground Control — Polyphony v3.26 + p1

Unofficial firmware customization · reference & install guide

## What this is

A browser patcher that applies **Polyphony v1** to official Endorphin.es Ground Control firmware v3.26. You supply the official `gc-*.hex`; the patcher changes only the intended bytes and writes a single `gc-*.hex` back to your card. Firmware is never uploaded.

## What Polyphony v1 changes

- **Clock priority.** Real-time bytes `0xF8`/`0xFA`/`0xFB`/`0xFC` move to the front of the MIDI event queue instead of waiting behind notes, so clock stays steady under load.
- **Polyphonic capture (tracks 1–3).** Every held note is stored per step, with its **real velocity**, instead of only the last note.
- **Stock-gate chord playback.** On playback each saved chord note is replayed through the real note emitter with the **same timing and gate as the primary note** — chords hold exactly like stock, and note-offs always pair (no stuck notes).

## Install

1. **Back up your entire microSD card** (projects, settings).
2. Download the official firmware `gc-*.hex` (v3.26) from endorphin.es.
3. Open `https://andrewdesena.github.io/ground-control-patcher/` in Chrome/Edge (https). Select the `.hex`, tick **Polyphony v1**, click **Build modded firmware**. Confirm the base image shows `✓ Ground Control v3.26`.
4. On the module hold **PROJ ~5 s** to mount the SD over USB. In the tool click **Connect GC SD card** and pick the card, then **Install to GC**.
5. The tool writes the patched file to the **card ROOT** under the **same `gc-*.hex` name**. *Do not rename it, and keep only one firmware `.hex` at the root.*
6. Press **PROJ ~5 s** again: the module reboots and flashes. Power-cycle when done.

> **The patcher creates a file; it does not flash the module.** The module's own bootloader flashes a `gc-*.hex` found at the card root. Keep the v2.x SD-recovery procedure handy for the first modded image.

## Recording & playing chords

Chords exist only for notes **recorded on this firmware** (they populate the per-step chord store). Record two or three notes on track 1 at one step, then play back — you should hear the chord. Projects recorded on stock firmware load as mono (their chord store is empty; a safe no-op).

## Verify

| | |
|---|---|
| Supported base | `v3.26 · canonical image SHA-256 e1db4815f8cf2983f832493aa8b2085586cb5ea10446e133774bdfc91bca9f52` |
| Build check | The tool re-parses its output and prints the **output SHA-256**; `tools/verify.py` proves a bundle moves **0** untouched bytes. |

## Limits (v1)

- Melodic tracks 1–3 only; assumes the default channels 1/2/3.
- Captured velocity falls back to 100 if a note-on velocity wasn't seen.
- Clock/polyphony behavior is experimental until hardware-verified.

## Return to stock

Put the original vendor `gc-*.hex` back at the card root and press **PROJ ~5 s**. The module flashes the original image.

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